

Nainsi Gupta

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Education

B.Tech. in Computer Science and Engineering

2022-2026

Vellore Institute of Technology | CGPA: 8.52/10

Technical Skills

Languages: C++, Python, JavaScript, SQL

Frameworks and Libraries: Scikit-learn, LangChain, TensorFlow, Keras, Numpy, Pandas, Seaborn, FastAPI, React.js, Node.js

Data Science & ML: Supervised/Unsupervised Learning, EDA, Model Evaluation, RAG, LLMs, Deep Learning, Model Deployment

Databases: MongoDB, MySQL, PostgreSQL, Vector Databases

Concepts: Data Structures & Algorithms, OOP, RESTful API Design, Software Engineering

Tools & Platforms: Git, GitHub, VS Code, Supabase, Vercel, Render, Jupyter Notebook, AWS, Power BI, Excel

Internship Experience

AI Engineer Internship | PearlThoughts (Remote)

Jan 2026 – Feb 2026

- Built a code intelligence & RAG-based system to index and query the ERPNext codebase using AST analysis, embeddings, and ChromaDB, enabling accurate, code-grounded understanding of large legacy systems.
- Implementing AI-assisted code intelligence tools aligned with a standardized RAG and context-generation pipeline, supporting analysis and modernization of large legacy codebases in a team setting.

Software Development Engineer Intern | Bluestock Fintech (Remote)

Jan 2025 – Feb 2025

- Developed a React.js and Node.js-based admin dashboard for IPO management, reducing data entry errors and improving the market disclosure process within the compliance department.
- Collaborated on a production-level IPO web application and REST API used by internal teams, utilizing GitHub for version control and following rigorous code review practices to maintain industry-standard quality.

Projects

Autonomous QA Agent (RAG-Powered Test Generation) | Python, FastAPI, LangChain, ChromaDB Sep, 2025

- Designed a Retrieval-Augmented Generation (RAG) pipeline using LangChain orchestration, HuggingFace embeddings, and Gemini LLM to generate software test cases from documentation and HTML structure.
- Implemented dual grounding with semantic retrieval (Chroma vector DB) and HTML parsing (BeautifulSoup), reducing hallucinations and improving element-level accuracy.
- Built a scalable FastAPI backend with vector indexing and contextual retrieval, along with a Streamlit frontend for interactive document ingestion, test generation, and Selenium script creation workflows.

Movie Recommender System | Python, Pandas, Scikit-learn, NLTK, Streamlit

 Jul, 2025

- Engineered an AI-powered movie recommender system using a hybrid approach that combines collaborative filtering and content-based filtering, delivering model accuracy of approximately 89.5%.
- Implemented both user-based and item-based similarity techniques using cosine similarity and TF-IDF vectorization to generate more relevant and personalized movie recommendations.
- Developed an interactive Streamlit dashboard for real-time personalized recommendations with intuitive visual analytics.

Traffic Forecasting using Graph Convolution - LSTM Model | TensorFlow, Keras, StellarGraph

 Mar, 2025

- Developed a hybrid GCN + LSTM model for traffic flow prediction, effectively capturing both spatial relationships between road networks and temporal traffic patterns.
- Performed data preprocessing and feature engineering using NumPy and Pandas, including train test splitting, normalization, and sliding window sequence generation for time series modeling.
- Represented the road network as a graph using distance-based adjacency matrices and evaluated performance using MAE and MASE, improving prediction accuracy through spatio-temporal modeling.

Achievements

Hackathon Recognition - Providence Leap Ideathon: Cleared the Second Round and recognized as a Next-Gen Thinker for designing an SMS/voice-enabled healthcare web app that improved data accuracy and enabled 3x wider reach in low-connectivity, underserved communities.

HP Power Lab 2.0: Cleared the Second Round, now in the Prototype Round, designing an AI-driven IoT Traceability System (Smart Drainer AI) to standardize used oil collection data and ensure full EPR compliance for the circular economy.

Certifications

- OCI 2025 Certified Generative AI Professional – Oracle University (Oct 2025)
- OCI 2025 Certified Data Science Professional – Oracle University (Oct 2025)